$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \mathrm{Pe} \nabla_x \cdot (v(\theta) f) + \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho)_\tau f) \\ f(t=0) &= f_0 \end{cases}$$

- What happens as $t \to \infty$?

- Convergent finite volume scheme (dW, Bruna & Schmidtchen, ESAIM. M2AN, 2025)

Ant analysis

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$\tau = 0$